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Problem

WHY IT MATTERS

- Modern generators modularize layers, but rarely modularize the entire generative model itself.
- Pixel-by-pixel images are high-dimensional and non-sequential; one long autoregressive sequence is prohibitively expensive.
- Latent tokenizers reduce cost but introduce reconstruction error and an information ceiling.

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Mathematics

RECURSIVE FACTORIZATION

- $\{x_{(i+1)}\} = g_i(x_i), N = k^n, n = \log_k(N)$
- Each level factorizes $p(x_1, \dots, x_N)$ into k conditional groups and delegates every group to the next-level generator.
- All levels are trained end-to-end, breadth-first, from the final raw-pixel cross-entropy loss.

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Core idea

NEW ABSTRACTION

- Treat an atomic generator as a reusable module and invoke the same kind of model recursively.
- Self-similar levels implement divide-and-conquer: images contain patches that are themselves images.
- Output count grows exponentially while recursive depth grows only linearly.

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Evidence

WHAT THE DATA SHOW

- ImageNet 64 x 64: FractalAR / FractalMAR reach 3.14 / 3.15 bits-per-dim NLL, beating prior likelihood models.
- At 256 x 256, finest-scale attention is about 4000 x cheaper than full next-scale attention.
- FractalMAR-H reaches FID 6.15, IS 348.9 and recall 0.46; it also supports inpainting and outpainting.

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Architecture

HOW IT WORKS

- The top generator splits a 256 x 256 image into 16 x 16 patches; child generators repeatedly subdivide them.
- The final lightweight transformer predicts RGB channels with 256-way cross-entropy.
- FractalAR uses raster-order causal attention; FractalMAR uses random-order bidirectional prediction.

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Meaning & limits

TAKEAWAY

- Raw-pixel generation is interpretable and avoids tokenizer reconstruction loss.
- The paradigm may extend to molecules, proteins and neural structures with intrinsic hierarchy.
- Generation remains autoregressive and scaling-heavy; larger models and broader domains are still open questions.

FRACTAL GENERATION

Fractal Generative Models

Li, Sun, Fan & He | 2025

Recursively compose whole generative models as modules, turning an intractable raw-pixel distribution into a hierarchy of manageable subproblems.

recursion

self-similarity

raw pixels